Figure 1.6: Two Common Ways to Represent Complex Numbers. In the Cartesian representation, x and y are the horizontal (real) and vertical (imaginary) components. In the polar representation, r is the radius, and θ is the angle made with the x axis. In each case, it takes two real numbers to represent a single complex number.

where x and y are real and $i^2 = -1$. Complex numbers can be added, multiplied, and divided by the standard rules of arithmetic. They can be visualized as points on the complex plane with coordinates x, y. They can also be represented in polar coordinates:

$$z = re^{i\theta} = r(\cos\theta + i\sin\theta).$$

Adding complex numbers is easy in component form: just add the components. Similarly, multiplying them is easy in their polar form: Simply multiply the radii and add the angles:

$$\left(r_1 e^{i\theta_1}\right)\left(r_2 e^{i\theta_2}\right) = \left(r_1 r_2\right) e^{i(\theta_1+\theta_2)}.$$

Every complex number z has a complex conjugate z^* that is obtained by simply reversing the sign of the imaginary part. If

$$z = x + iy = re^{i\theta},$$

then

$$z^* = x - iy = re^{-i\theta}.$$

Multiplying a complex number and its conjugate always gives a positive real result:

$$z^* z = r^2.$$